Toward a Multi-Source Framework for Forecasting Urban Real Estate Trends in Ho Chi Minh City

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***Abstract*—Urban real estate markets are influenced by heterogeneous information sources such as property transactions, administrative land records, and macroeconomic indicators. The fragmentation of these data sources, together with complex temporal dynamics, makes reliable housing price forecasting a challenging task. Traditional statistical models such as ARIMA have long been used for economic time-series forecasting due to their interpretability and ability to capture temporal dependencies [1]. More recently, deep learning approaches, particularly Long Short-Term Memory (LSTM) networks, have demonstrated strong performance in modeling nonlinear temporal patterns in economic and financial data [3].**

This study proposes a structured multi-source data integration framework for urban real estate trend analysis, organizing heterogeneous urban datasets into a unified analytical pipeline and evaluating multiple forecasting approaches under a consistent experimental setting. The empirical evaluation is conducted on real-world data collected from three western districts of Ho Chi Minh City. After data cleaning and preprocessing, 5,171 transaction records are aggregated into a monthly time series consisting of 47 observations representing the average housing price per square meter. Three forecasting approaches—ARIMA, Prophet, and LSTM—are evaluated under a unified experimental pipeline.

Experimental results show that the Prophet model achieves the best forecasting performance on the test set, outperforming both ARIMA and LSTM in terms of MAE, RMSE, and MAPE. The findings suggest that for relatively short urban real estate time series, decomposable statistical forecasting models can provide more stable and accurate predictions than deep learning approaches.

The proposed framework demonstrates the feasibility of using real-world urban datasets for data-driven analysis and forecasting of real estate market dynamics.

The source code and processed datasets used in this study are publicly available at: https://github.com/HaPXV/Urban-Real-Estate-Forecasting.

***Index Terms*—Real Estate Forecasting, Time-Series Prediction, Multi-Source Urban Data, ARIMA, Prophet, LSTM**

I. Introduction

Urban real estate markets play an important role in economic development, urban planning, and investment decision-making. Housing prices reflect not only supply and demand conditions in the property market, but also broader economic factors such as credit availability, ownership transfer activity, and investment flows. As a result, accurate forecasting of real estate price dynamics has become an important research problem in both urban economics and data-driven urban analytics.

Forecasting urban real estate markets remains challenging for two main reasons. First, relevant information is often distributed across heterogeneous sources such as transaction records, land registration systems, and macroeconomic indicators. Second, many practical urban datasets are relatively short, noisy, and difficult to model using a single forecasting paradigm. These challenges are particularly important in emerging urban markets, where real-world datasets are often fragmented and less standardized than benchmark datasets used in prior studies.

Traditional time-series models such as the AutoRegressive Integrated Moving Average (ARIMA) have long been used in economic forecasting because of their interpretability and ability to capture temporal dependencies in historical observations [1]. More recently, machine learning and deep learning approaches, especially Long Short-Term Memory (LSTM) networks, have been adopted for housing price prediction [3], [4], [6]. In parallel, the integration of heterogeneous urban datasets has attracted increasing attention as a way to improve the representation of market conditions [5].

Despite these advances, two limitations remain in the existing literature. First, many real estate forecasting studies still rely on a single data source, such as listings or transaction records, which may not capture the multi-dimensional nature of urban market dynamics. Second, direct comparisons between classical statistical models and deep learning approaches under the same real-world experimental setting remain limited, especially for short time-series datasets from emerging markets.

To address these limitations, this paper proposes a structured forecasting framework for urban real estate trend analysis that integrates heterogeneous urban datasets from Ho Chi Minh City. The framework is designed to accommodate multi-source data while providing systematic model comparison under a unified experimental pipeline, even when full multivariate integration is constrained by data availability in emerging urban markets. The framework organizes transaction records, administrative land registration data, and macroeconomic indicators into a unified monthly analytical structure and evaluates three representative forecasting models: ARIMA, Prophet, and LSTM.

The main contributions of this study are summarized as follows:

- We propose a multi-source data integration framework that organizes heterogeneous urban datasets — including transaction records, administrative land registration data, and macroeconomic indicators — into a unified analytical pipeline for urban real estate analysis. Empirical evaluation is conducted on the transaction-based price series as a controlled first step, with the framework designed to support full multivariate extension as broader data coverage becomes available.
- We provide a systematic comparison between classical statistical models and deep learning approaches for relatively short urban time-series datasets.
- We present an empirical case study using real-world real estate data from Ho Chi Minh City, providing new evidence for forecasting real estate dynamics in emerging urban markets.

II. Related Work

Research on real estate price forecasting has been widely explored in the fields of urban analytics, econometrics, and machine learning. Existing studies generally fall into three main directions: statistical time-series modeling, machine learning approaches, and spatial or multi-source urban data analysis.

A. Statistical Time-Series Forecasting

Classical statistical models such as the AutoRegressive Integrated Moving Average (ARIMA) have long been used for economic and financial forecasting due to their ability to capture temporal dependencies in historical observations [1]. ARIMA-based approaches remain popular in real estate forecasting because of their interpretability and relatively strong performance on small or medium-sized time-series datasets.

More recently, decomposable time-series models such as Prophet have gained attention in business and economic forecasting applications [2]. Prophet models time series as a combination of trend, seasonal, and residual components, allowing it to handle missing observations and structural changes more robustly than many traditional models. These properties make Prophet particularly suitable for real-world economic datasets that often contain noise and irregularities.

B. Machine Learning Approaches

Machine learning methods have also been widely explored for housing price prediction. Neural network models can capture nonlinear relationships among property attributes, spatial characteristics, and macroeconomic indicators [4]. In particular, Long Short-Term Memory (LSTM) networks have been applied to time-series forecasting because they can model long-term temporal dependencies in sequential data [3], [6].

Recent studies have also explored more advanced machine learning frameworks for real estate analysis and valuation, including graph-based representation learning models such as MugRep [7]. These approaches attempt to capture complex spatial and relational structures in urban environments. However, many deep learning models typically require large training datasets to achieve stable performance, which may not always be available in real-world urban datasets.

C. Spatial and Multi-Source Urban Data Analysis

Another important research direction focuses on integrating heterogeneous urban datasets. Geographic Information Systems (GIS) and spatial econometric models have been used to analyze spatial heterogeneity in housing prices [8]. These approaches emphasize the role of geographic location and neighborhood characteristics in real estate valuation.

Recent studies have also explored the use of multi-source urban data, including transaction records, geospatial information, and socio-economic indicators, to improve predictive modeling of housing markets [5]. The integration of multiple data sources can provide a richer representation of market dynamics and improve the explanatory power of forecasting models.

D. Research Gap

Despite these advances, most studies rely on single data sources and lack systematic comparisons between classical and deep learning approaches under unified real-world settings — particularly for short time-series datasets in emerging markets. This study addresses both gaps using real-world data from Ho Chi Minh City.

III. Data and Preprocessing

A. Data Sources

To analyze urban real estate market dynamics, this study integrates multiple real-world datasets collected from different sources. These datasets represent complementary perspectives of the urban real estate market, including market transactions, administrative land registration activities, and macroeconomic indicators.

The primary dataset (Dataset A) consists of real estate transaction records collected from online property listings and transaction-related sources in three western districts of Ho Chi Minh City: Tan Binh, Tan Phu, and District 12. After data cleaning and preprocessing, the dataset contains 5,171 valid transaction records. Each record includes attributes such as property price, property area, property type, construction level, location characteristics (e.g., frontage or alley), and transaction timestamp. These records provide direct information about market price formation and transaction behavior in the study area.

The second dataset (Dataset B) contains land registration records obtained from administrative sources. These records reflect official ownership transfer activities such as property purchases, transfers, and inheritance registration events. Compared with listing-based transaction data, administrative registration records provide additional insights into the intensity of real estate transaction activities and ownership changes.

The third dataset (Dataset C) consists of macroeconomic indicators that may influence real estate market dynamics. These indicators include the central bank interest rate, gold price, and the VN-Index stock market index. Macroeconomic variables are incorporated to represent broader economic conditions that may affect investment behavior and housing market activity [9].

TABLE I
SUMMARY OF DATASETS USED IN THE STUDY

Dataset	Description	Data Type	Time Period
Dataset A	Real estate transaction records from Tan Binh, Tan Phu, and District 12 (5,171 cleaned records)	Market data	2022–2025
Dataset B	Land registration records reflecting ownership transfer activities	Administrative data	2024–2025
Dataset C	Macroeconomic indicators including interest rate, gold price, and VN-Index	Economic data	2022–2025

Table I summarizes the main characteristics of the datasets used in this study.

B. Data Cleaning and Integration

Because the datasets originate from heterogeneous sources, several preprocessing steps were required before modeling. First, invalid or incomplete records were removed to ensure data quality. Records with missing key attributes such as transaction price, property area, or transaction timestamp were excluded from the dataset.

Second, price values were normalized into a unified metric of price per square meter, enabling meaningful comparison across properties of different sizes. Statistical filtering techniques were applied to detect extreme outliers that could distort the aggregated time-series representation.

Third, the datasets were aligned along a common temporal dimension. Since the forecasting models operate on time-series data, all records were aggregated into monthly intervals. Transaction data were summarized into monthly statistics including the average price per square meter and the number of transactions. Administrative registration records were aggregated into monthly counts of ownership transfer activities, while macroeconomic indicators were aligned with the same monthly timeline.

The processed datasets were then merged using the monthly time index to form a unified analytical dataset. In the current experiments, the forecasting models operate on the transaction-based price series derived from Dataset A. Dataset B covers a different administrative district (Thu Duc) and a shorter time span (2024–2025), limiting its direct integration as a covariate in the current setting. Dataset C macroeconomic indicators are incorporated into the framework design and reserved for multivariate extension in future work. This phased approach reflects practical data constraints common in emerging urban markets, where administrative and economic records are often fragmented across agencies and geographic boundaries. This design allows the study to evaluate forecasting behavior under a controlled transaction-based setting before extending to richer multivariate configurations.

C. Time-Series Construction

After preprocessing and data integration, the real estate data were transformed into time-series representations suitable for forecasting models. Two primary variables were constructed.

The first variable is the average real estate price per square meter, which represents the overall price trend in the study area. The second variable represents market activity and is defined as the number of transactions or registration events per month.

Monthly aggregation was used to reduce short-term fluctuations in individual transactions and to generate a stable time-series signal for forecasting. The resulting dataset forms a structured time-series representation in which the housing price series is potentially influenced by auxiliary signals from administrative and macroeconomic datasets.

This representation enables the application of both classical statistical forecasting models and machine learning approaches, which are described in the following section.

IV. FORECASTING FRAMEWORK AND MODELS

A. Forecasting Framework

The proposed forecasting framework aims to analyze and predict urban real estate market trends using time-series representations derived from heterogeneous urban datasets. The framework follows a structured pipeline consisting of three main stages: data integration, model training, and forecasting evaluation.

First, heterogeneous datasets including transaction records, administrative land registration data, and macroeconomic indicators are integrated and aggregated into a monthly analytical structure, as described in Section III. The primary forecasting target is the average real estate price per square meter, while auxiliary signals provide contextual information about market activity and economic conditions.

Second, multiple forecasting models are trained using historical observations from the constructed monthly time series. In this study, three representative forecasting approaches are considered: ARIMA, Prophet, and Long Short-Term Memory (LSTM). These models represent different methodological paradigms including classical statistical forecasting, decomposable time-series modeling, and deep learning approaches.

Finally, the trained models are evaluated using standard forecasting accuracy metrics to compare predictive performance across methods. Figure 1 illustrates the overall pipeline of the proposed forecasting framework.

B. ARIMA Model

The AutoRegressive Integrated Moving Average (ARIMA) model is a classical statistical method widely used for economic time-series forecasting [1]. ARIMA models capture temporal dependencies through autoregressive and moving-average components combined with differencing to handle non-stationary time series.

Due to its interpretability and effectiveness for relatively small datasets, ARIMA is commonly used as a baseline model in forecasting studies. In this work, ARIMA is applied to model the temporal dynamics of the real estate price time series.

C. Prophet Model

Prophet is a decomposable forecasting model designed for business and economic time-series prediction [2]. The model

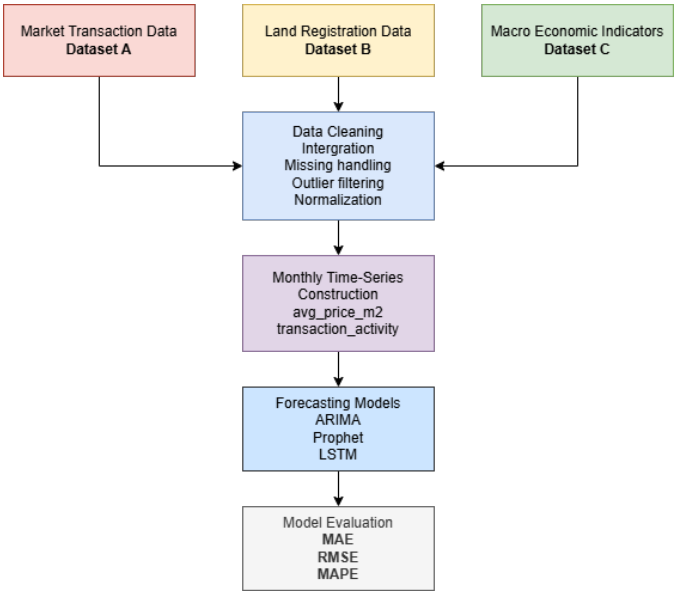

Fig. 1. Proposed forecasting pipeline integrating heterogeneous urban datasets and time-series forecasting models.

represents a time series as the combination of trend, seasonal, and residual components, allowing it to handle missing values and structural changes effectively.

Because of its robustness and ease of implementation, Prophet has been widely adopted for real-world forecasting tasks. In this study, Prophet is applied to model the housing price time series and evaluate its forecasting performance in the urban real estate context.

The Prophet model was configured with default hyperparameters (changepoint_prior_scale = 0.05, seasonality_mode = 'additive'), which were found to be appropriate for the relatively smooth trend of the housing price series.

D. LSTM Model

Long Short-Term Memory (LSTM) networks are a type of recurrent neural network designed for sequential data modeling [3]. LSTM models are capable of capturing nonlinear temporal dependencies and have been widely applied in time-series forecasting tasks.

In this study, the LSTM model is trained on the constructed real estate time-series dataset to represent a deep learning forecasting approach.

E. Training Strategy

To ensure a fair comparison among forecasting models, all models are trained and evaluated under the same experimental protocol. The dataset is divided into training and testing sets according to temporal order to avoid information leakage.

Historical observations are used for model training, while the most recent observations are reserved for testing. For machine learning models such as LSTM, input sequences are generated using sliding windows over the time-series data, enabling the model to learn temporal relationships between past observations and future values.

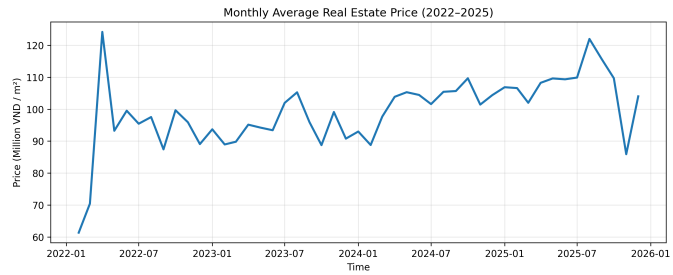

Fig. 2. Monthly average real estate price (million VND/m^2) in the study area from 2022–2025.

F. Evaluation Metrics

Forecasting performance is evaluated using three commonly used metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{1}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2}$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{3}$$

These metrics provide complementary measures of forecasting accuracy and enable systematic comparison among forecasting models.

V. Experimental Results

A. Experimental Setup

The experiments were conducted using the monthly aggregated real estate transaction dataset derived from Dataset A. After data preprocessing and filtering procedures described in Section III, the dataset contains 5,171 valid transaction records collected from three districts in Ho Chi Minh City: Tan Binh, Tan Phu, and District 12.

The transaction records were aggregated into monthly statistics, resulting in a time series consisting of 47 observations covering the period from 2022 to 2025.

To evaluate forecasting performance, the time series was divided into training and testing subsets according to temporal order. The first 37 months were used for model training, while the remaining 10 months were reserved for testing. This temporal split reflects realistic forecasting conditions where future observations must be predicted based only on past information.

Figure 2 illustrates the monthly average real estate price series constructed from the transaction dataset.

Three forecasting models were evaluated in this study:

- ARIMA(3,1,3), representing classical statistical time-series forecasting
- Prophet, a decomposable forecasting model widely used in economic prediction

TABLE II
FORECASTING PERFORMANCE COMPARISON

Model	MAE	RMSE	MAPE (%)
ARIMA (3,1,3)	7.68	9.57	7.32
Prophet	**6.49**	**9.02**	**6.46**
LSTM	8.86	10.78	8.46

- Long Short-Term Memory (LSTM), representing deep learning approaches

The ARIMA(3,1,3) configuration was selected through a grid-search procedure on the training set, where candidate models with $p, q \in \{0, 1, 2, 3\}$ and both $d = 0$ and $d = 1$ were examined using AIC/BIC criteria and then validated by out-of-sample forecasting errors on the test set. The final ARIMA(3,1,3) model was retained because it achieved the best overall trade-off between information criteria and forecasting stability among the evaluated candidate models.

The LSTM model consists of a single LSTM layer with 32 hidden units followed by a dense output layer. The model was trained using the Adam optimizer with a learning rate of 0.001 for 100 epochs. A sliding window of length six months was used to construct input sequences.

B. Stationarity Analysis

Before training the forecasting models, the stationarity of the housing price time series was examined using the Augmented Dickey-Fuller (ADF) test. The ADF test produced a statistic of -5.4796 with a p-value below 0.001, indicating strong evidence of stationarity in the processed time series used for ARIMA modeling.

This result supports the suitability of ARIMA-based forecasting and confirms that differencing-based temporal modeling is appropriate for the housing price series.

C. Model Performance Comparison

Table II presents the forecasting performance of the evaluated models on the test dataset.

Among the three primary forecasting models evaluated, Prophet produced the lowest MAE, RMSE, and MAPE values, indicating more accurate predictions than both ARIMA and LSTM on the test set.

The ARIMA model captured the general temporal structure of the housing price series but produced slightly larger forecasting errors than Prophet. The LSTM model achieved the lowest performance among the evaluated approaches, which is likely due to the relatively small size of the dataset consisting of only 47 monthly observations.

D. Forecast Visualization

Figure 3 compares the actual housing price series with the Prophet forecasts on the test dataset.

The figure shows that the Prophet model captures the overall trend of the housing price series and produces relatively small prediction errors across most months in the testing period.

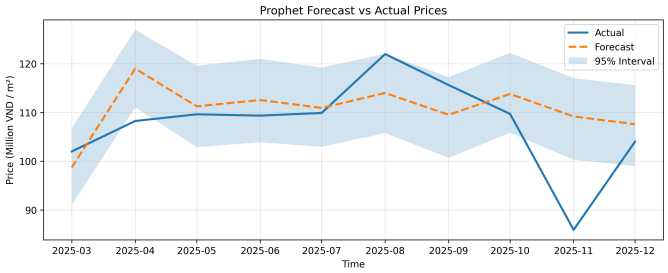

Fig. 3. Comparison between actual prices and Prophet forecasts on the test set.

E. Robustness Across Different Train/Test Splits

To evaluate the robustness of forecasting performance, additional experiments were conducted using multiple temporal splits: 32/15, 35/12, and 37/10 months for training and testing respectively. These splits were selected to examine how forecasting accuracy changes when the amount of historical training data varies.

Table III reports the Mean Absolute Percentage Error (MAPE) obtained under different split configurations.

TABLE III
ROBUSTNESS EVALUATION ACROSS DIFFERENT TRAIN/TEST SPLITS

Train/Test Split	ARIMA (3,1,3)	Prophet
32/15	11.76%	8.51%
35/12	6.42%	8.19%
37/10	7.32%	6.46%

The notably higher ARIMA error at the 32/15 split (11.76%) suggests that the ARIMA(3,1,3) configuration, which was selected based on the full training set, may be over-parameterized for shorter training windows. Prophet's additive decomposition structure appears more robust to reduced training lengths in this dataset.

Although forecasting accuracy varies across splits due to the limited length of the time series, the Prophet model maintains relatively stable performance and achieves the best result under the primary experimental split (37/10).

F. Comparison with Baseline Methods

In addition to ARIMA and Prophet, two simple baseline forecasting strategies were evaluated: naive forecasting and a moving average model with a window size of three months.

Table IV summarizes the forecasting accuracy of all evaluated methods.

TABLE IV
COMPARISON WITH SIMPLE FORECASTING BASELINES

Model	MAE	RMSE	MAPE (%)
Naive	6.61	9.02	6.41
Moving Average (3)	6.87	9.12	6.62
ARIMA (3,1,3)	7.68	9.57	7.32
Prophet	**6.49**	**9.02**	**6.46**
LSTM	8.86	10.78	8.46

The results indicate that simple baselines such as naive forecasting can achieve competitive performance due to the relatively smooth trend of the housing price series. However,

Prophet provides more consistent forecasting accuracy across experiments.

G. Discussion on Model Sensitivity

The experimental results reveal several important characteristics of the evaluated forecasting models.

First, the performance of the models is influenced by the relatively small size of the available time-series dataset. The dataset consists of only 47 monthly observations, which limits the amount of historical information available for training deep learning models. As a result, the LSTM model does not outperform the statistical forecasting approaches in this study.

Second, the forecasting performance varies slightly across different training and testing splits, as shown in Table III. This variation reflects the sensitivity of time-series models to the amount of historical data used for training. Nevertheless, the Prophet model demonstrates relatively stable performance across different experimental settings.

Third, the comparison with simple baseline models indicates that the real estate price series exhibits a relatively smooth temporal trend. In such situations, naive forecasting strategies may produce competitive results. However, Prophet consistently provides more stable predictions across experiments and better captures long-term trend changes in the time series.

These observations highlight the importance of selecting forecasting models that are appropriate for the size and characteristics of the available dataset. In the context of limited historical observations, decomposable statistical models such as Prophet may offer more robust performance than deep learning approaches.

It is worth noting that the Naive forecasting baseline achieves a MAPE of 6.41% on the primary test split, which is marginally lower than Prophet's 6.46%. However, Prophet outperforms Naive in terms of MAE (6.49 vs. 6.61 million VND/m^2), and the two models produce identical RMSE values. More importantly, the robustness evaluation in Table III demonstrates that Prophet maintains consistent performance across multiple train/test splits, whereas the Naive baseline was evaluated only under the primary experimental configuration. These results suggest that Prophet provides more reliable generalization behavior, which is a critical property for practical forecasting applications.

VI. Conclusion

This paper presented a data-driven framework for analyzing and forecasting urban real estate market trends using heterogeneous urban datasets collected from Ho Chi Minh City. The study integrates transaction records, administrative land data, and macroeconomic indicators to construct a time-series representation of housing market dynamics.

Three forecasting approaches were evaluated, including ARIMA, Prophet, and Long Short-Term Memory (LSTM). Experimental results show that the Prophet model achieved the best forecasting performance on the constructed dataset, producing lower forecasting errors compared with both ARIMA and LSTM models. Additional experiments using baseline models and multiple train–test splits further confirm the robustness of the forecasting results.

The findings suggest that decomposable statistical forecasting models such as Prophet can provide stable and reliable predictions for urban real estate markets when historical data is limited, particularly in emerging-market settings with short real-world time series.

However, the study has several limitations that suggest directions for future work. First, the available time-series dataset contains only 47 monthly observations, which restricts the ability of deep learning models such as LSTM to fully exploit temporal patterns — a finding consistent with prior literature on small-sample time-series forecasting. Second, the geographic scope of auxiliary datasets is constrained by practical data availability: administrative land registration records (Dataset B) cover a different district and a shorter time window, while macroeconomic indicators (Dataset C) require longer overlap periods to serve as stable covariates. These constraints reflect the fragmented nature of urban data infrastructure in emerging markets such as Vietnam, and motivate the phased framework design adopted in this study. Full multivariate integration across all three data sources remains an important direction for future research as data coverage expands.

Future work will pursue three directions. First, extending the transaction dataset with longer temporal coverage and broader geographic scope will enable more robust training of deep learning models. Second, incorporating spatial property features and point-of-interest data will support property-level valuation beyond aggregated market indicators. Third, full integration of administrative registration records and macroeconomic indicators as multivariate inputs — once sufficient temporal alignment is achieved — will validate the complete multi-source pipeline proposed in this framework.

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